

Integration Test

May 8, 2026

Problem 1: Integral of $e^x \cos(x)$

To evaluate the integral $\int e^x \cos(x) dx$, we will use integration by parts, which states $\int u dv = uv - \int v du$. We will apply this method twice.

Let $I = \int e^x \cos(x) dx$.

For the first application of integration by parts, let: $u = \cos(x) \implies du = -\sin(x) dx$ $dv = e^x dx \implies v = e^x$

Applying the integration by parts formula:

$$I = e^x \cos(x) - \int e^x (-\sin(x)) dx$$

$$I = e^x \cos(x) + \int e^x \sin(x) dx$$

Now, we need to evaluate the integral $\int e^x \sin(x) dx$. Let's call this J . We apply integration by parts again to J . For this second application, let: $u = \sin(x) \implies du = \cos(x) dx$ $dv = e^x dx \implies v = e^x$

Applying the integration by parts formula to J :

$$J = e^x \sin(x) - \int e^x \cos(x) dx$$

Notice that the integral on the right side, $\int e^x \cos(x) dx$, is our original integral I . So, we can write J in terms of I :

$$J = e^x \sin(x) - I$$

Now substitute this expression for J back into the equation for I :

$$I = e^x \cos(x) + (e^x \sin(x) - I)$$

$$I = e^x \cos(x) + e^x \sin(x) - I$$

Now, we solve this equation for I :

$$I + I = e^x \cos(x) + e^x \sin(x)$$

$$2I = e^x (\cos(x) + \sin(x))$$

$$I = \frac{e^x}{2} (\cos(x) + \sin(x))$$

Finally, we add the constant of integration C :

$$I = \frac{e^x}{2} (\cos(x) + \sin(x)) + C$$

The final answer is $\boxed{\frac{e^x}{2} (\cos(x) + \sin(x)) + C}$.